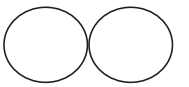
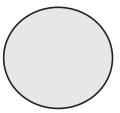
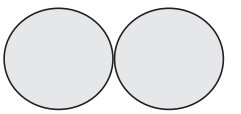
Joining elements (1)

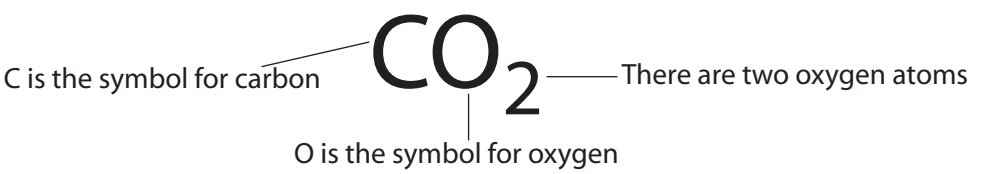
In some **elements**, such as oxygen, the atoms pair up. A particle that is made up of more than one atom is called a **molecule**. We can show this using the chemical symbols for the atoms involved. For example, oxygen molecules can be represented as O_2. The number in the **formula** shows how many atoms of the type there are in the molecule.

We can represent molecules by drawing coloured circles to show the atoms. When we draw a molecule we draw the atoms of that type touching, to show that they are chemically joined together.

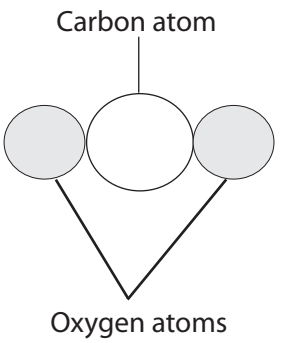
Element	Formula for molecule	Picture of molecule
oxygen – a non-metal gas 	O_2	 oxygen gas
nitrogen – a non-metal gas 	N_2	 nitrogen gas

When two or more atoms from different elements join together, they form a **compound**. The formula for a compound shows the types of elements that are in the compound and the ratio (how many atoms of each element there are).

Carbon dioxide is a compound made up of carbon and oxygen. It has the formula:



This formula tells us that a molecule of carbon dioxide is always made up of one carbon atom and two oxygen atoms. We don't need to add a one after carbon. A chemical formula only includes a number after an element symbol if there is more than one atom of that element.



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Joining elements (2)

Read the Resource sheet 'Joining elements (1)', then answer the following questions.

☞ **1** Use the words in the word bank to complete the table.

Word	Description
atom	Tiny p _____ that cannot be b _____ down into s _____ substances
element	Made up of only o _____ type of a _____
compound	Made up of t _____ or more e _____ joined together
molecule	Two or more a _____ joined together. If the atoms are the s _____ it is a m _____ of an element. If the atoms are d _____, it is a molecule of a c _____.

Word bank
atoms
broken
compound
two
particle
simpler
different
same
one
elements
atom
molecule

2 Decide whether each of the following statements is true or false.

a An element is made up of only one type of atom. _____

b A compound can be made of atoms that are the same or different. _____

c A molecule is always made up of two or more different types of atoms. _____

d We use a formula to show the type of elements and the number of atoms of each element in a molecule. _____

e Nitrogen dioxide has the formula NO_2. This tells us that there are two elements present, nitrogen and oxygen, and that one molecule contains one nitrogen atom and two oxygen atoms. _____

3 Circle the element symbols and underline the formulas of compounds in the box below.

SO_2	Cu	Zn	NaCl	Cl	H_2O	K

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